

Def. A pointwise linear map

$$\varphi: \bigcup_{p \in \mathbb{R}^3} T_p \mathbb{R}^3 \rightarrow \mathbb{R}$$

is called a 1-form on $\mathbb{R}^3$. The pointwise linear means: for fixed $p \in \mathbb{R}^3$,

$$\varphi_p: T_p \mathbb{R}^3 \rightarrow \mathbb{R}; V_p \mapsto \varphi(V_p)$$

is linear functional of $T_p \mathbb{R}^3$.

Define the sum of 1-forms φ_1 and φ_2 as

$$(\varphi_1 + \varphi_2)(V_p) \stackrel{\text{def}}{=} \varphi_1(V_p) + \varphi_2(V_p) \text{ for any tangent vector } V_p.$$

And $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ and 1-form φ ,

$$(f\varphi)(V_p) \stackrel{\text{def}}{=} f(p) \cdot \varphi(V_p)$$

Def. Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a differentiable function. The 1-form

$$df(V_p) \stackrel{\text{def}}{=} V_p[f]$$

is called the differential of f .

Lemma. For fixed $p \in \mathbb{R}^3$, $\{dx_i \mid i=1,2,3\}$ is the dual basis for $T_p \mathbb{R}^3$ corresponding to $\{e_1(p), e_2(p), e_3(p)\}$.

That is, given 1-form φ ,

$$\varphi = \sum_{i=1}^3 \varphi(e_i) \cdot dx_i.$$

Moreover, for differentiable f ,

$$df = \sum_{i=1}^3 \frac{\partial f}{\partial x_i} dx_i.$$

Proof. Let $p \in \mathbb{R}^3$ be fixed. Then $dx_i(e_j(p)) = \delta_{ij}$. Therefore, given $V = (V_1, V_2, V_3)$,

$$\varphi_p(V_p) = \varphi_p(V_1 \cdot e_1(p) + V_2 \cdot e_2(p) + V_3 \cdot e_3(p)) = \sum_{i=1}^3 \varphi_p(e_i(p)) \cdot V_i = \sum_{i=1}^3 \varphi_p(e_i(p)) \cdot dx_i(V_p)$$

Particular, $df(e_i(p)) = e_i(p)[f] = \frac{\partial f}{\partial x_i}$.

Computation of forms.

Let $f(x, y, z) = (x^2 - 1)y + (y^2 + 2)z$. Then,

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$= 2xy dx + (x^2 + 2yz - 1) dy + (y^2 + 2) dz.$$

So $V_p[f] = df(V_p) = 2p_1 p_2 v_1 + (p_1^2 + 2p_2 p_3 - 1) v_2 + (p_2^2 + 2) v_3,$
proof.